

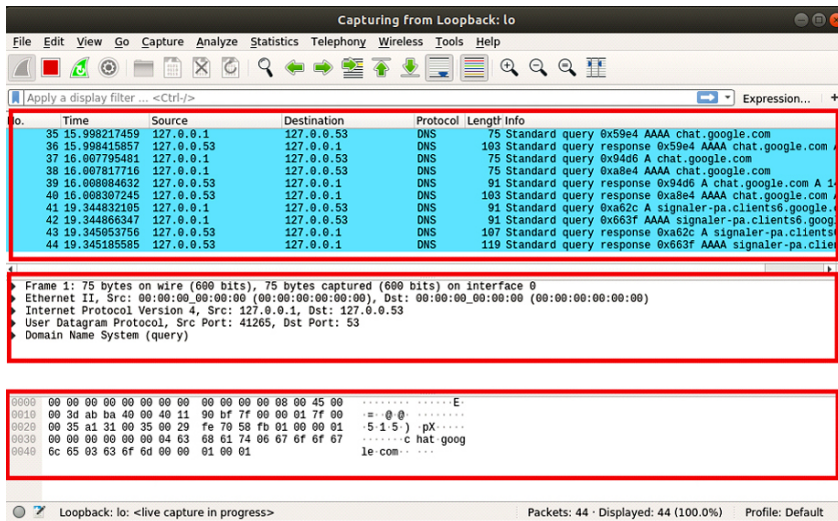


Figure 7: Packets captured from the Loopback:lo interface

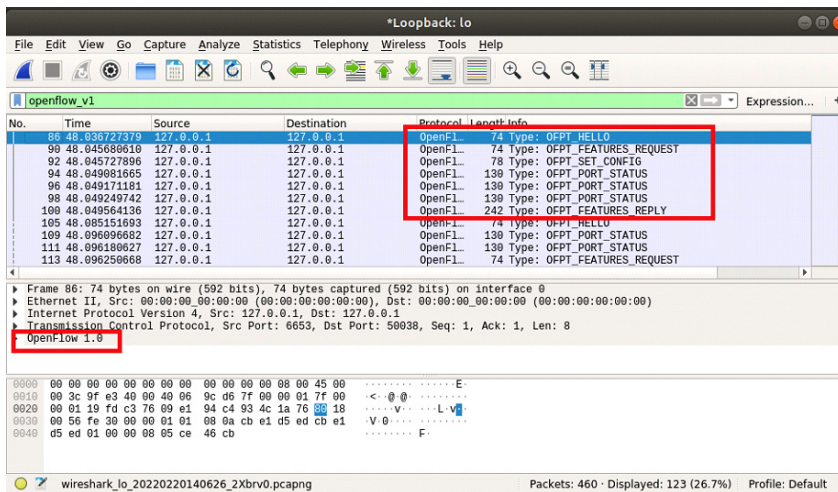


Figure 8: Capturing OpenFlow packets using Wireshark

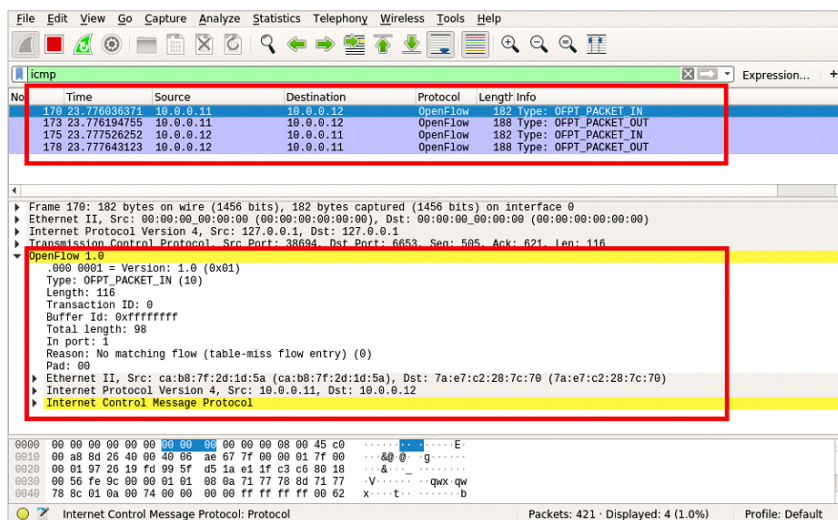


Figure 9: OpenFlow packets between host h1 and host h2

being transmitted between the controller and the data plane (switch) in SDN. The steps to trace the OpenFlow packet are given below.

Step 1: Open the terminal in Linux and run the following command to launch Wireshark:

```
#sudo wireshark &
```

Step 2: The Wireshark window (Figure 6) shows the list of interfaces that Wireshark can capture network traffic on. Click on 'Loopback: lo' and then start capturing the packets.

Step 3: When the packet capturing starts, the Wireshark window (as shown in Figure 7) will be divided into three frames. The first part displays the captured packets, which includes their number, time, source and destination IP addresses, protocol name, length, and information of the captured packet. The second part contains detailed information about every captured packet. The third part contains the real data that is captured in the packet.

Step 4: We can apply filters to display a specific set of captured packets. To show OpenFlow packets only, write *openflow_v1* in the Wireshark filter text box, and press *Enter*. When you press the *Enter* key, it won't show any packets in the packet capturing section because we haven't run the created topology and started the controller that exchanges OpenFlow packets with the data plane devices.

Step 5: To run the custom topology and for starting the controller, use the following command:

```
#sudo python mytopo.py
```

The above command will start the controller and activate the OpenFlow application. We will see the number of OpenFlow packets displayed in the Wireshark window, as shown in Figure 8.